ASSIGNMENT-(1) [DBMS]

[Section-A]

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Section-(A)

Q.1 What is DBMS? Why is it required?

↳ A Database management system (DBMS) is software used to store, manage and retrieve data efficiently.

Required because:-

- Data security
- Data Consistency
- Reduced redundancy
- Easy data access
- Backup and recovery
- Multi-user access

Q.2 Why is DBMS better than Traditional File system?

- Reduces redundancy or data duplication
- Avoid inconsistency
- Provides security
- Supports Concurrency
- Backup and recovery
- Data Sharing

Q. What is Data Abstraction? Explain levels.

- Data abstraction means hiding internal database details.

Levels:

1. Physical level – Describes how data is actually stored in the database.
2. Logical level – Describes what data is stored and the relationships between them.
3. View level – Describes how different users see specific parts of database

Q. Three - Schema architecture and importance.

1. External level – Different user view of database
2. Conceptual level – Overall logical structure of the entire database.
3. Internal level – Physical storage details of the database

Importance :-

- Data independence
- Structured database design

Q. Define data independence.

↳ Data independence is the ability to modify schema without affecting other levels.

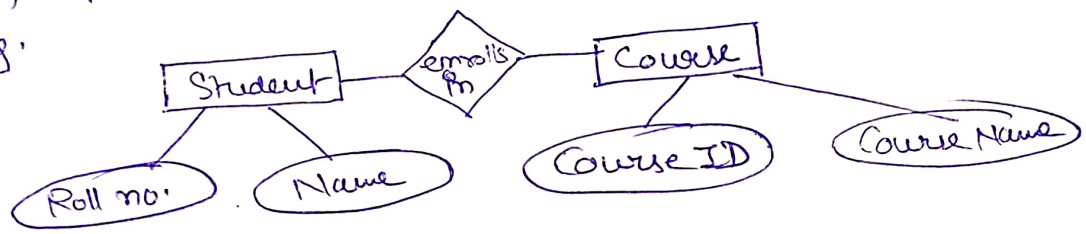
Types :-

- Physical Data independence
- logical Data independence

Q. What is ER Model :-

↳ ER Model represents entities, attributes and relationships graphically.

eg.



Q. Integrity Constraints

⇒ Rules that maintain data accuracy, and consistency in database

Types :-

- Primary Key - Uniquely identifies each record and cannot be NULL.
- Foreign Key - Ensures values match a primary key in another table.
- Unique - Ensures all values in a column are different.
- Not Null - Ensures a column cannot have NULL values
- Check - Ensures column values satisfy a specified condition.

[Section-B]

Q. Draw three-schema architecture and explain each level.

External level (User views / Applications) ⇒ Shows how different users view specific parts of database.

|

Conceptual level (Logical Database Schema) ⇒ Describes the overall logical structure of ~~entities~~ entire database.

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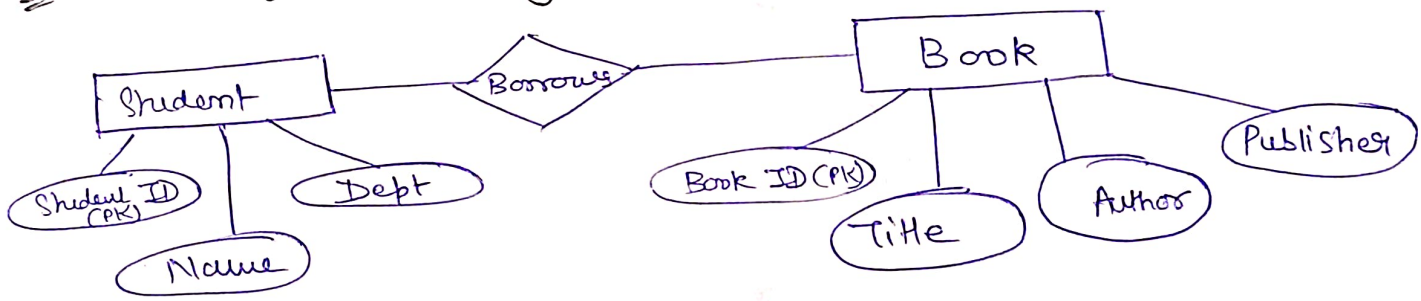
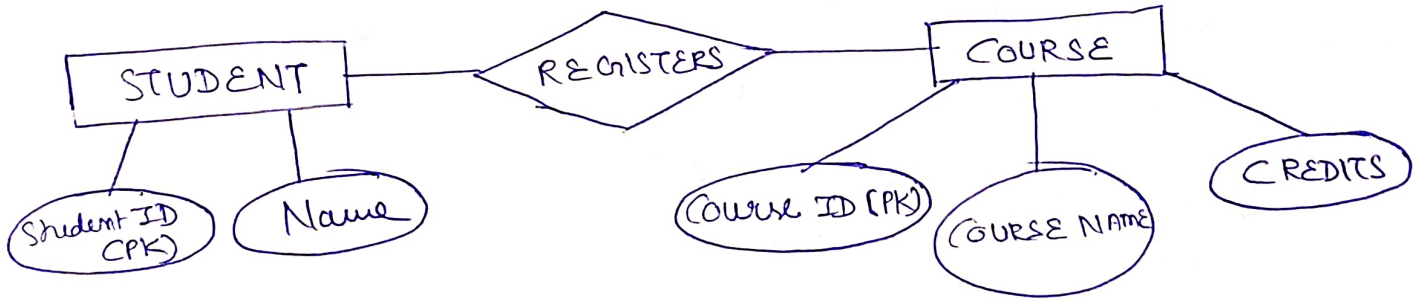
Internal level (Physical storage level) ⇒ Describes how data is physically stored in memory.

Q. ER Diagram - Library management system.



Q. ER Diagram - Student Course Registration



[Section-C]

Q. A hospital wants to store patient, doctor, and appointment data. Identify entities, attributes, and relationships.

- Entities :-
  - Patient
  - Doctor
  - Appointment

Attributes

- Patient (Patient_ID, Name, ~~Age~~ Age)
- Doctor (Doctor_ID, Name, Specialization)
- Appointment (Appointment_ID, Date, Time)

Relationship

- Doctor treats Patients
- Patient books Appointment

Q.2 An online shopping platform stores customer, product and order data. Suggest keys and constraints

- Primary keys ⇒
  - Cust_ID
  - Product_ID
  - Order_ID

- Constraints ⇒
  - Foreign Keys
  - NOT NULL
  - UNIQUE
  - CHECK (price > 0)

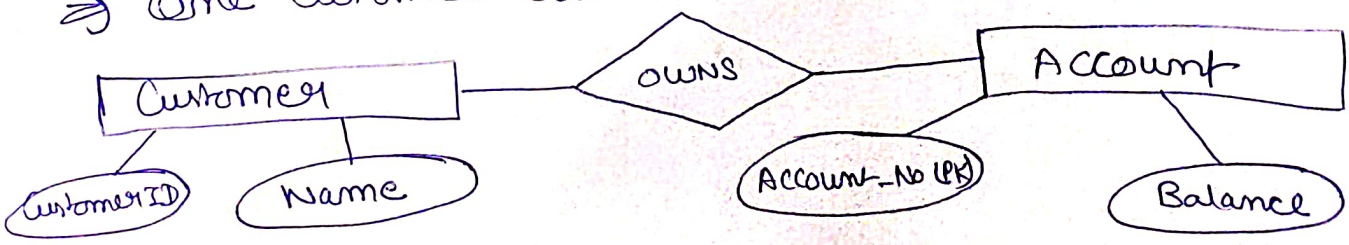
Q.3 A university stores student records. Explain how data abstraction helps different users.

⇒
- View level - Student access
- Logical View - Admin management
- Physical level - Data storage.

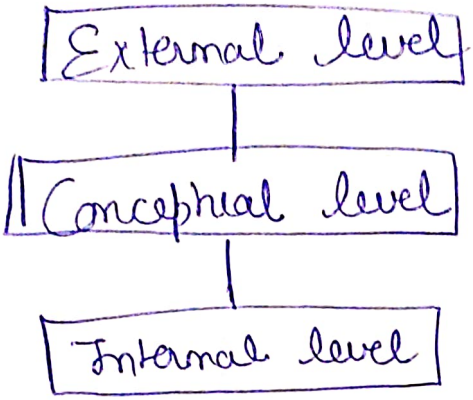
[Section-D] ODONI

Q. Draw a simple ER diagram for any real-life system and explain one key design decision.

⇒ One Customer can have multiple accounts.



Q. Draw the three-schema architecture and mention one benefit.

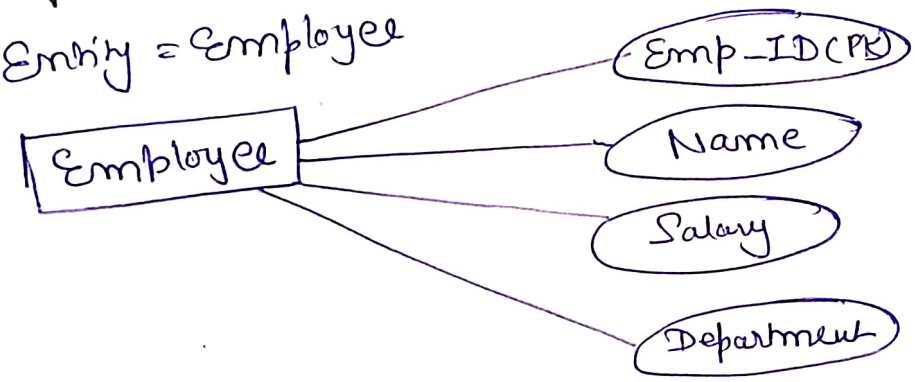


[Benefit]

↳ Provides data independence

Q. Draw an entity with attributes and identify its primary key.

⇒ Entity = Employee



Primary key - Emp_ID

~~[Section E]~~